

Algorithms

What programs are made of

How do you cook rice?

What is a computer

A thing that does stuff

For the purpose of this course

We will define computers as

A machine that takes in *instructions* and *completes* those instructions in order

What are instructions

The term you'll start being familiar with is

Algorithms

An algorithm is a *recipe*, a set of instructions to accomplish a task.

A *program* is simply a set of algorithms

How computers execute algorithms

Computers can **only**

- perform simple arithmetic and data movement, and
- remember results and place them somewhere

Any and *all* complexity that you see in modern computers is a result of these simple operations

minecraft

.....

Computers will only do what you tell it to do

Example using a simplified computer

Memory, Control Unit, Arithmetic Logic Unit, Input, Output

Where the initial instructions are

```
1  Add 3456, 3457
2  Store 3458
3  Add 3361, 34462
4  Store 3456
5  Compare, 3456, 3458
6  Print
```

Where the initial memory state is

address	value
3456	3
3457	4
3458	
3459	True
3460	False
3461	5

Robot

8x8 grid,

- move forward,
- turn 90 right

Maze